

## Session 2: Choosing research project

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# Overview I

- 1 Selecting research project
- 2 Where to find data
- 3 Break
- 4 Quarto project
- 5 Tasks for the week
- 6 Any questions?

# Learning goals of today

- Selecting a good research project

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- Selecting a good research project
- Finding data for research project
- Starting the project in R

## Selecting research project

# Why a research project

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- Passive learning less effective
- Active learning by doing improves learning outcomes

# Learning by doing

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- I want to guide you through the process
- Empowering you to continue learning

# Good research projects

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## Good research projects

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- **A**chievable
- **R**elevant
- **T**imely

# Specific

- Less specific: Has population growth increased?

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- More specific: Has population growth across countries increased the past 10 years?

# Measurable

- Less measurable: How has sentiment on inflation changed?

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- More measurable: How have people's responses to inflation questions in standard surveys changed over time?

# Achievable

- Less achievable: How have consumption patterns of Germans versus non-Germans changed over the past 40 years using G-SOEP data?



# Achievable

- Less achievable: How have consumption patterns of Germans versus non-Germans changed over the past 40 years using G-SOEP data?
- More achievable: What are consumption patterns of Germans versus non-Germans using G-SOEP data?

# Relevant

- Less relevant: How often do people in Germany use their car?

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- More relevant: How has car use changed in light of the recent increases in fuel costs due to the war on Iran?

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- Less timely: How has the Spanish flue pandemic from 1918-20 changed population at local levels using historical data?
- More timely: How has the Covid-19 pandemic affected population across the ages using UN population estimates?

## Where to find data

# Types of data

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  - Administrative data

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- Ideally random selection allows for inference of population parameters using sample estimates
- Very expensive if done well

## Survey data: Pros and Cons

### PRO

- Include wide array of questions

### CONTRA

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- Can be quite complex
- Often more difficult to gain access

## Survey data: Examples

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<https://www.europeansocialsurvey.org/>
- World Value Survey: <https://www.worldvaluessurvey.org/>
- For a range of surveys (in Germany) check out GESIS:  
<https://www.gesis.org/angebot/daten-finden-und-abrufen/ueberblick>

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- Data often collected more consistently (in Global North)
- Be careful of changes in definition of underlying properties (!)

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- Often requires more data wrangling to combine
- May have data gaps or breaks in temporal data

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- OECD:  
<https://www.oecd.org/en/data/datasets.html?orderBy=mostRelevant&page=0&facetTags=oecd-languages%3Aen>



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- Example: <https://www.geeksforgeeks.org/data-science/world-bank-dataset-in-r/>



Break

## Quarto project

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- Projects are self-containing environments for us to work in
- Idea is that everything you need to work on your project is in one place
- The project becomes (more) independent from the environment it is executed in
- Enhances portability to other machines (your own or others)

# Starting new project

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- I will demonstrate in RStudio

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- Makes sure that research is reproducible

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- If your way of doing things changes the results of your calculations each time you do it that is very bad
- Reproducibility and transparency also (!!!) for your future self



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- Way to refer to your files on your machine may depend on operating system used
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- In Windows, most files are stored on C:\
- File location can be copied from top bar in Windows Explorer
  - Just make sure to use the correct or /
- Use clearly identifiable folder names, such as \data or \output

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- Output files (such as PDF, HTML, PNG etc.)

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- Notes

## Example header

```
#-----  
#  
# Name: Data preparation  
# Purpose: Preparation of OAA2 data analysis  
# Data: OAA2 data, collected the in May and June 2024  
# Author: Heiner Salomon (heiner.salomon@uni-bremen.de)  
# Date created: 12/05/24  
#  
# Notes -----  
#  
# Preparation -----
```





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  - format: pdf

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- Set up the research project in Quarto
- Write the progress report by Friday at 15:00

## Reading for next week

- Gentzkow, M. and Shapiro, JM. (2014): Code and Data for the Social Sciences: A Practitioner's Guide <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>

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- Gentzkow, M. and Shapiro, JM. (2014): Code and Data for the Social Sciences: A Practitioner's Guide <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>
- We will discuss those ideas at the beginning of next week's class

Any questions?